Lagrangian Mechanics and the Principle of Least Action

PHYS 6801 — Classical Mechanics (Example)

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1 Principle of Least Action

Definition 1 (Action)**.** *The* action *S associated with a path $q(t)$ is defined as the time integral of the Lagrangian L along that path:*

$$S = \int L\big(q(t), \dot{q}(t), t\big)\, dt. \tag{1}$$

We seek paths for which the action is stationary under variations $\delta q(t)$ of the trajectory. Requiring the first variation of the action to vanish gives the condition

$$\delta S = 0. \tag{2}$$

This variational principle selects the physical path among all paths with fixed endpoints.

2 Euler–Lagrange Equation

Imposing (2) on (1) yields the Euler–Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0. \tag{3}$$